Appendix A
Uncertainties

Introduction – What Is Uncertainty?

DEFINITIONS

These will help you understand some words commonly used in everyday language that have a different meaning in the scientific lab.

Data - (or 'results') a series of measurements taken during an experiment. Each measurement should be stated with its uncertainty.

Uncertainty - (terms also used are **errors** and **confidence limits**) the amount of *accuracy* you can state your measurement due to the precision of the equipment you are using. The uncertainty is usually stated as a 'plus or minus' value e.g. ±0.1 using *one* significant figure.

Resolution - (or accuracy or precision) the smallest measurement you can make with a device determined by the scale used. For example, a ruler with mm gradings has better resolution than one with only cm gradings.

Error Analysis - working out the uncertainties of all data measurements and subsequent calculations to determine the accuracy of your final result.

Mistake - (or 'human error') is usually due to incorrect use of equipment, faulty equipment, incorrect analysis or assumptions. These are to be avoided and corrected rather than included in your final error analysis.

Systematic Errors - occur with every measurement due to a consistent mistake or error in the experimental setup, procedure or analysis. These are particularly hard to spot as they skew all the data, not just one measurement. Reaction time with a stop watch is an example.

Results - the final calculation of all your data determined by mathematical and/or graphical analysis using the physics theories. This MUST also include an uncertainty to demonstrate how accurate the result is and how well it compares with the theory.

The following pages will explain the processes and mathematics for working out the uncertainties and how to do the error analysis during your experiments. Please read carefully, complete the practice problems before your first prac.

Error Analysis is our set of tools for dealing with experimental uncertainties. The aim is not to eliminate uncertainties – unfortunately, that's usually impossible. But by figuring out how best to estimate our uncertainties, we will produce practical and useful real-world results.

A Golden Rule

Ultimately, your analysis of uncertainties is based on your own estimates and judgment. The methods outlined below are guidelines, **not** absolute laws; every experimental situation will be unique. The key is always to think carefully about what you are doing, to ensure that your analysis is as reasonable and as sensible as possible.

Uncertainties in Measurements

The most basic uncertainties are those due to measurements. Whenever you measure a physical quantity, that measurement has an uncertainty. This is true even if we assume that the equipment is perfectly accurate and correct.

There are different sources of uncertainty detailed below. Sometimes only one sort of uncertainty will be relevant or significant; in other situations, there may be multiple sources of uncertainty to deal with. (If you are unsure which uncertainty is more relevant, you should probably choose the largest one.) Recognising which uncertainties are appropriate in which situations can be confusing at first. When in doubt, talk to your demonstrator.

If a **quantity** is labelled 'x', then its **uncertainty** is labelled 'Δx'.

So when we state a **numerical result**, we write '$x \pm \Delta x$'. E.g. 3.4 ± 0.1 mm

NOTE: In the context of experimental results and uncertainties, Δ always means 'the uncertainty in', (not 'change in').

Δx is called the **absolute uncertainty** of x.

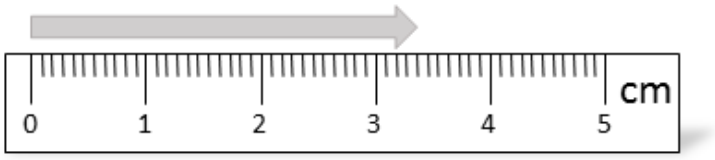
$\frac{\Delta x}{x}$ is the **proportional uncertainty** of x.

$\frac{\Delta x}{x} \times 100$ is the **percentage uncertainty** of x.

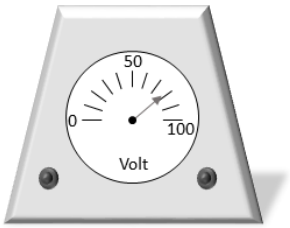
Determine the proportional AND percentage uncertainty of 3.4 ± 0.1 mm

Taking a Measurement: The 'Reading Error'

When reading a measurement from a device you will have a **reading error** due to the **resolution** of the measurement scale on the device. This may be the marks on a ruler, needle on a dial or display on a digital readout.



This 'reading error' uncertainty is generally estimated to be plus-or-minus half the smallest division of measurement. So, a ruler which is marked in millimetres can only be used to measure distances to the nearest millimetre. For example, using the ruler above we might measure the length of the arrow to be 34 mm long. We should state this result as 34.0 ± 0.5 mm. The uncertainty is 0.5 mm due to the reading error: if the result had been more than 34.5, then it would be read as 35, and if less than 33.5, then it would be read as 33 mm. Therefore, the appropriate range is 34.0 ± 0.5 mm.



State the two measurements to the left and their uncertainties. (Don't forget the units.)

Significant Figures and Rounding Off

The precision of an experimental result is implied by the number of digits recorded in the result although generally the uncertainty should be quoted specifically as well. The number of ***significant figures*** in a result is defined as follows:

1. The leftmost nonzero digit is the most significant digit.
2. If there is no decimal point, the rightmost nonzero digit is the least significant digit
3. If there is a decimal point, the rightmost nonzero digit is the least significant digit, even if it is a 0.
4. All digits between the least and most significant digits are counted as significant digits.

For example, the following numbers each have four significant digits: 1234, 123,400, 123.4, 1001, 1000., 10.10, 0.0001010, 100.0. If there is no decimal point, there are ambiguities when the rightmost digit is 0. Thus, the number 1010 is considered to have only three significant digits even though the last digit might be physically significant. To avoid ambiguity, it is better to supply decimal points or to write such numbers in ***scientific notation***, that is, as an argument in decimal notation multiplied by the appropriate power of 10. Thus, our example of 1010 would be written as 1010. or $1.010 * 10^3$ if all four digits are significant.

Taking Several Measurements: Error from Averages

Sometimes we are measuring something which we know is stable and easily controlled (like the length of a piece of string). These sorts of measurements may only need to be made once, and so the reading error is the uncertainty.

However, much of the time the measurements of the quantity can vary, giving you a slightly different answer every time. The variability of this measurement forms the basis of the uncertainty in this case. By finding the **average** of many repeat measurements and the variation of these measurements we can determine the final result and its uncertainty.

Note that the more trials or repeat measurements you take the average will become more accurate and you will reduce the likelihood of mistakes. But the range of measurements are also likely to increase. If you are taking a large number of measurements (more than 12) you should use the **standard deviation** to determine the uncertainty.

HOW TO DETERMINE THE UNCERTAINTY OF AN AVERAGE (Up to 12 data points).

Mean or Average $\bar{x}$: $\bar{x} = \frac{1}{N}\sum_1^N \bar{x}_i$

$$\bar{x} = \frac{153.2+153.6+152.8+153.0}{4} = 153.2 \text{ mm}$$

Range ($x_{\max} - x_{\min}$): $153.6 - 152.8 = 0.8$ mm

Uncertainty (half range): $\Delta x = \frac{0.8}{2} = 0.4$ mm

Final Result: $x = 153.2 \pm 0.4$ mm

*If the half range uncertainty (± 0.4) is larger than the reading error (± 0.1), use the half range uncertainty in your final result. If the reading error was larger, you would use ± 0.1. When in doubt, use the *largest* uncertainty.

STANDARD DEVIATION

If there are lots of measurements, then the half-range error becomes less accurate. So we use the **standard deviation** to determine the spread of the data.

$$\sigma_x = \sqrt{\frac{1}{N-1}\sum_1^N (x_i - \bar{x})^2}$$

with the measurement results x_i and the mean or average $\bar{x}$. The pre-factor $\frac{1}{N-1}$ is called Bessel's correction and accommodates for the bias in the estimation of the population variance (more information can be found on the Wikipedia page; https://en.wikipedia.org/wiki/Variance).

The next Figure shows the amount of variation or dispersion of a set of data values. 68.2 % of the measured data points x_i are in the window of $\bar{x} \pm \sigma$.

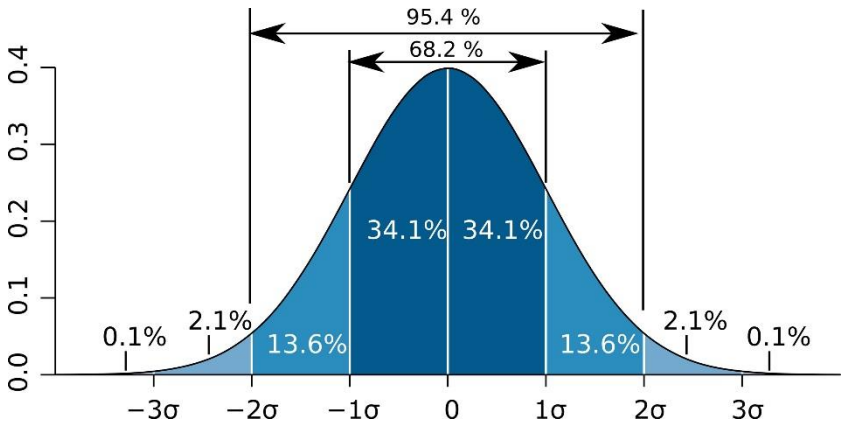


Fig. Standard deviation plot for a normal distribution.

If it is appropriate to use these techniques, they will be detailed in the laboratory exercise notes.

Find the half range uncertainty and the standard deviation for the set of data below;
23.5 ± 0.2, 25.2 ± 0.2, 24.8 ± 0.2, 23.9 ± 0.2, 26.0 ± 0.2, 22.7 ± 0.2

Minimising Errors

Uncertainty Analysis (see above) deals with the statistical uncertainties due to experimental measurement. However, basic statistical uncertainty analysis assumes that the actual measurements are still 'correct'. In real life, this is not always the case – equipment can be faulty, procedure can be incorrect, assumptions can be false, etc. It is important to understand the ways in which experimental errors can occur, and to learn to recognise and minimise them.

You will encounter mistakes and systematic errors during your experiments. It is important to look out for these, understand how they came about and show how you fixed them. Use the strategies below to help you do this.

How To Minimise Error

There are various general methods by which you can minimise the error and uncertainty associated with a measurement. The most important thing is simply to **think** about what you are doing. Don't just blindly follow directions! You should critically assess your experimental method as you go along. Does what you are doing make sense? If not, why not?

TRIAL RUNS

Whenever possible, you should do a brief trial run with your equipment before you take your 'actual' measurements. This will allow you to familiarise yourself with the equipment, and it provides an opportunity to discover potential sources of error sooner. If problems do occur, you will save time by altering your procedure before (instead of during) the actual experiment.

If you make a mistake (or you suspect that you might have), you should repeat your measurements to check your results. If you are unsure that you are using your equipment correctly, re-read the directions in the manual; if you are still unsure, talk to your demonstrator.

Don't try to cover up your mistakes! Discuss them in your logbook and detail what was changed to solve the problem. You will not lose marks for catching a mistake and fixing it, but you will lose marks if you try to cover it up or ignore it.

EQUIPMENT ASSESSMENT

You need to decide whether the equipment to be used is accurate enough for the task at hand. Can you (or should you) calibrate it? Should alternative equipment be used instead?

CAREFUL USE OF EQUIPMENT

For example:

- Meters with needle indicators should be tapped lightly, to check that the indicator is not sticking
- Check that the 'zero' settings on instruments are actually registering as 'zero'
- Arrange apparatus so that it cannot be easily knocked or bumped
- Connect electrical circuits and all electronic components securely

CONSISTENT PROCEDURE

Follow standard procedures carefully. When you change procedure or devise your own techniques, write out your new method in point form and follow it closely. If you are consistent and methodical in your approach, you will minimise the variability that comes from haphazard equipment use.

CHECKS AND CROSS-CHECKS

You shouldn't just record data: you need to think about what the results mean, and check that your answers make sense. Consider the following:

- Can you check your results while you are taking them, to see if they are consistent?
- Can you and your partner both take readings, and check each other's work as you go?
- Can you draw a graph as your results come in, to see if they are making sense?
- Can you already guess, roughly, what the results should be? Are they doing what you expect?

ANALYSIS

Think about how you are analysing your data to ensure you are using the correct mathematics and physics.

- Only round off numbers at the end of a calculation.
- Make sure your assumptions are correct or appropriate for the experimental conditions.
- Make sure the theory you use is applicable for the experimental conditions.

PRESENTING THE FINAL RESULT

When presenting the final result of your experiment, you should consider the uncertainty of the final result too. In later years you will determine the uncertainty of your final result mathematically and take into account all measurements.

In first year, we would like you to estimate the uncertainty of your results due to the **largest** measurement uncertainty. You can do this by determining the largest %uncertainty of your measurements and using the same percentage for your results.

If you took measurements of 25.3 ± 0.5 mm (or 2% uncertainty) and 38 ± 2 V (or 5% uncertainty) the second measurement is likely to impact on your overall results the most.

Your final result was 1502 V/m with a 5% uncertainty or 1502 ± 75 V/m.

NOW you can discuss the validity of your result by comparing to the expected value.

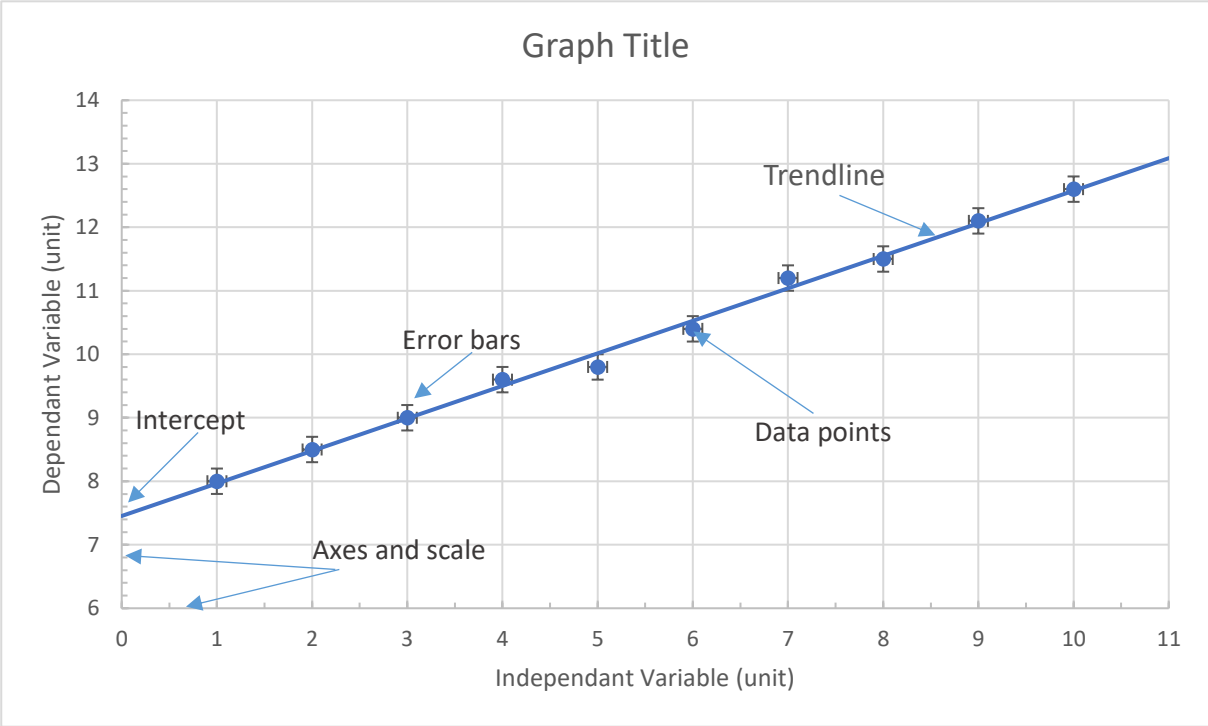
Additional information can be found in:

https://en.wikipedia.org/wiki/Propagation_of_uncertainty
http://spiff.rit.edu/classes/phys273/uncert/uncert.html
https://www.dit.ie/media/physics/documents/GPG11.pdf
https://en.wikipedia.org/wiki/Variance
http://en.wikipedia.org/wiki/Error_propagation

Appendix B
Graphs and How to Use Them

Graphs are a great way to represent and analyse data. Trends and relationships between variables become clear when the data are graphed in a meaningful way. It is important to learn both how to create useful graphs and how to extract information from them.

Plotting Graphs – Key Points and Elements



- **Labels** – always label both *axes* and the *graph* itself. Be sure to include UNITS eg: Length (m), Time (x 10^2) (s).

- **Axes** – each graph will have a vertical and horizontal axis
    - **independent** variable - this is the quantity that you control, and that you have chosen the initial values of – e.g. the force that you apply to an object. *Horizontal axis*
    - **dependent** variable - this is the quantity which varies due to the independent variable – e.g. the acceleration that results from the force you apply. Usually, the dependent variable is the thing that we are investigating. *Vertical Axis*
    - **scale -** use a scale that displays your data in the best way. The range should spread the data evenly across the graph which may mean that you will need to exclude the zero point of an axis.

- **Data** - Plot data points clearly, **include error bars** (see page A-8) to display the uncertainty values of each point.

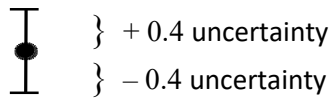
- **Trendline** – a line of best fit through the data points (see page A-9).
    - **Intercept** – point on axis (horizontal and/or vertical) where trendline crosses.

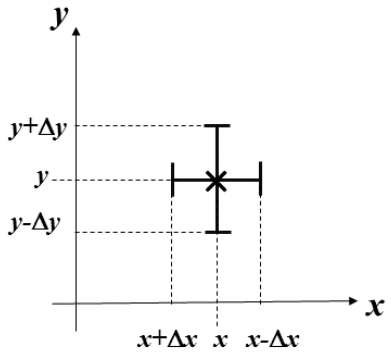
Error Bars — Graphing Uncertainties

You must always show the uncertainty values of your results on graphs. Uncertainties are displayed on graphs using **error bars**, which show the 'plus-or-minus' values on either side of a data point.

For example: a distance measurement 'x' equal to 153.2 ± 0.4 mm can be displayed below:

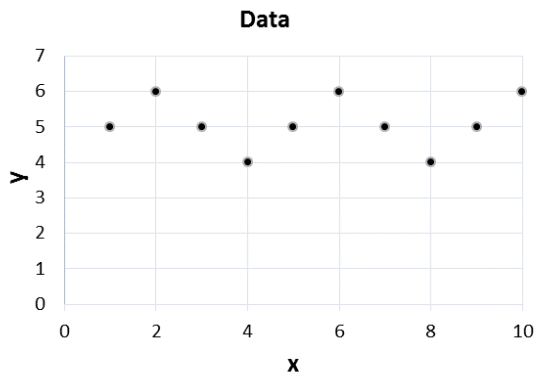
$$x + \Delta x = 153.2 + 0.4 = \quad 153.6$$
$$x = \quad 153.2$$
$$x - \Delta x = 153.2 - 0.4 = \quad 152.8$$



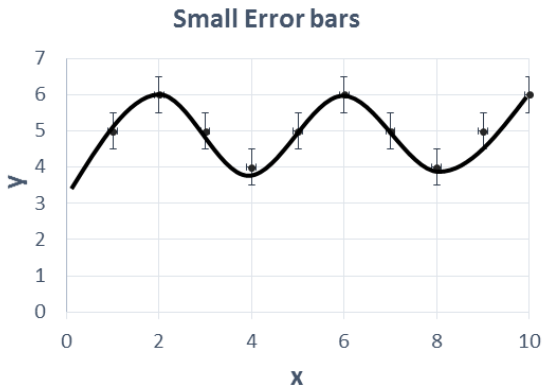


Data points will often have errors due to both the independent and dependant variables, so we should include both **horizontal** and **vertical** error bars. In this way, the error bars will mark out a region of uncertainty in which we expect the 'actual' data point (x,y) to be.

We need error bars to be able to see the true relationships between data. For instance, without error bars we might obtain a graph like this:

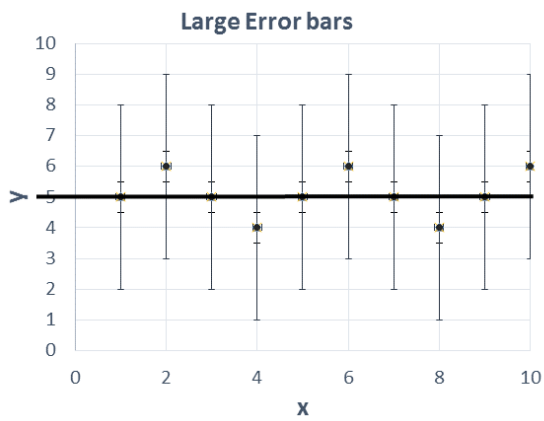


It appears as if these points could be joined together to fit a sine wave. However, this may not actually be the case. If the uncertainties for x and y are both small (e.g. ~5%), then the error bars look something like this:



As you can see, a sinusoidal curve fits within these error bars quite well, and so it is reasonable to conclude that this relation is probably a sine wave. However, if the uncertainties are larger (e.g. ~50%), then the error bars look something like this:

In this case, a straight line would fit the data just as well as a curve. It may be a sine wave relation, but the uncertainties are too large for us to be sure.

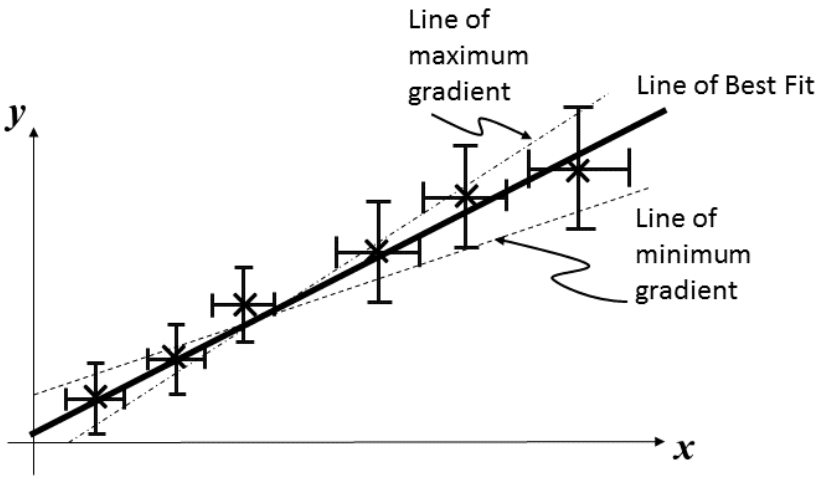


Straight Lines

In experimental work, we often want to look for a **linear relationship** between two physical quantities. If a graph of y versus x produces a straight line, then we may conclude that the relationship between x and y is **linear**. This means that the quantities are related by a linear equation (i.e. an equation of the form $y = mx + c$).

Unfortunately, data points will rarely form a perfectly straight line on a graph. In these cases, we need to find the **line of best fit**.

The following graph represents a typical example:



- Draw a line of best fit with a ruler that passes inside all (or most) of the error bars.
- Determine the gradient and y-intercept of your line and present as $y=mx+c$
- Draw two more lines that represent the minimum and maximum gradients (m_{min} and m_{max}) the data can have, again passing through most if not all the error bars. Find their gradients.
- The uncertainty of the gradient can be determined by the difference of the min and max gradients. i.e. gradient $= m \pm \Delta\left(\frac{m_{max-m_{min}}}{2}\right)$

Examine your data carefully: there may be one or two points which are obviously very different to the others. If you believe that these differences are mistakes or due to some experimental error, then you may decide to ignore them when constructing your line of best fit. (Also be sure to investigate how and why this happened and include a comment!)

Linearising your Data - converting Curves to Straight Lines

A graph will often look more like a *curve* than a straight line. Unfortunately, it is difficult to sketch a reliable curve from scattered data points, and it is even more difficult to extract an obvious mathematical relation from a curve. However, if we already have a theoretical relation for our data, we can sometimes use this to convert our curve graph into a straight-line graph. A straight-line graph can be used to extract a linear relationship between the data (see **Straight Lines**, above).

By converting to a linear graph, we make it much easier to analyse the theoretical relationship between the data. It also makes it easier to extract the information corresponding to the gradient 'm' and the axis intercept 'c'.

Linear relationships are also simpler to extrapolate beyond the measured data. It is easy to see how a straight line will continue across the graph; extrapolating a curve is more difficult.

Converting to a Straight Line — Example

The period $\boldsymbol{T}$ of circular motion of an object of mass $\boldsymbol{M}$ depends on the centripetal force $\boldsymbol{F}$ causing the circular motion, and the radius $\boldsymbol{r}$ of the motion, given by the following equation:

$$F = \frac{4\pi^2 Mr}{T^2}$$

When $\boldsymbol{F}$ versus $\boldsymbol{T}$ is graphed, we will expect to see a curve corresponding to the above equation.

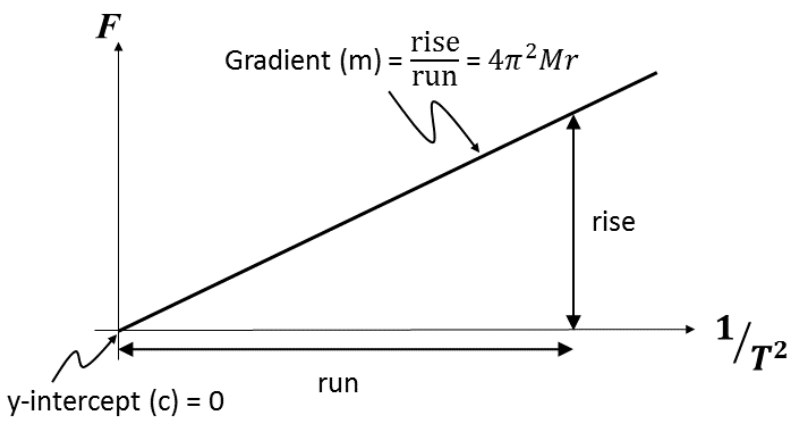
We can turn this curve into a straight line by rewriting the above equation into a **linear form**.

$$F = \frac{4\pi^2 Mr}{T^2}$$

$$F = \left(4\pi^2 Mr\right)\left(\frac{1}{T^2}\right) + 0$$

$$y = \quad (m) \qquad (x) \quad + c$$

The equation is now in a linear form (i.e. $\boldsymbol{y} = m\boldsymbol{x} + c$). By graphing $\boldsymbol{F}$ versus ($1/\boldsymbol{T}^2$) instead of $\boldsymbol{F}$ versus $\boldsymbol{T}$, we should create a linear graph instead of a curve graph. Instead of looking for a curve to check that the data matches the equation, we can instead look for a straight line.



$\boldsymbol{F}$ corresponds to $\boldsymbol{y}$	:	vertical axis
$\frac{1}{T^2}$ corresponds to $\boldsymbol{x}$	:	horizontal axis
$4\pi^2 Mr$ corresponds to m	:	gradient
$\boldsymbol{F}$ axis intercept corresponds to c	:	in this case, zero

Using Excel for Equations and Graphs

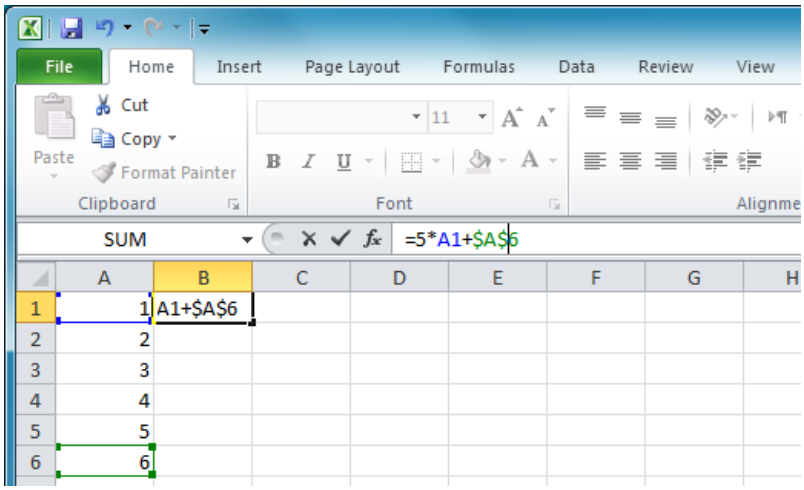
Often you will be creating your graph with Excel which you can print out and paste into your log book.

Data

Enter your data into a spreadsheet in Excel. Include a column for errors if you have them for both your variables. Use clear labels and units for your table. You may also have to use equations in your spreadsheet.

Equations

Sometimes you may need to use an equation to manipulate your data before you graph it or produce the final answer.

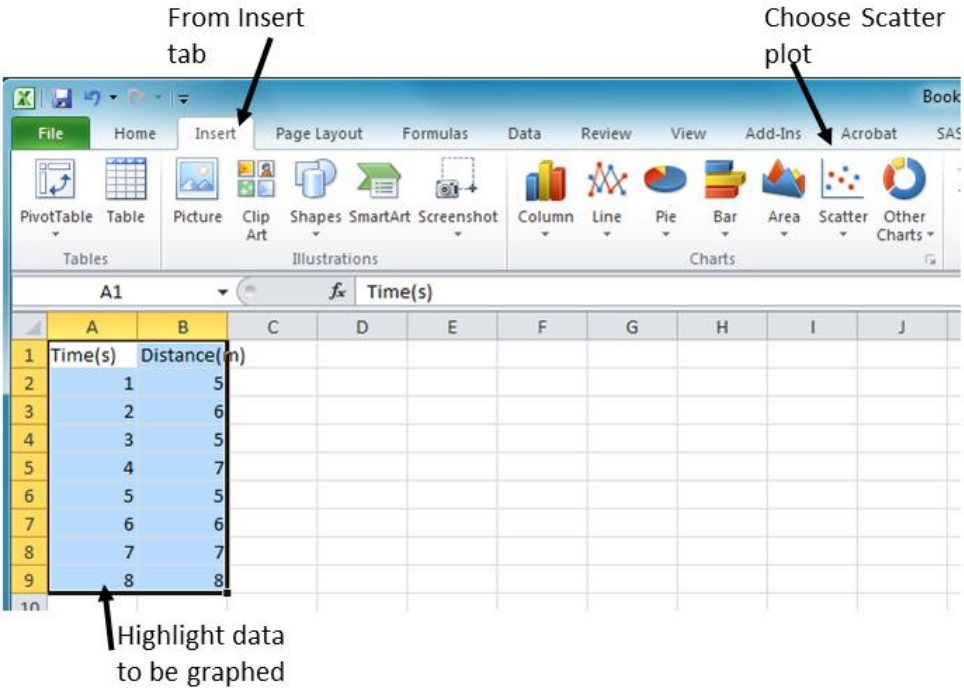


- Click in the cell you want to put the new data in. At the Equation Line type '=' followed by the equation you wish to use. If you are using data from another cell in your equation – click that cell instead of writing in the number.
- Using '$' forces that column or row to stay the same. So **A6** will always have the value of **6**, whereas **A1** will change to **A2** if you drag the equation down. You do this by clicking and dragging the little black square on the bottom right corner.
- Here are some of the more common commands you may use to create an equation;

operation	**command**	**example**
x^2	^2	=A1^2
xb	*	=A1*B1
$\frac{x}{b}$	/	=A1/A2
$\sum x$	SUM(number1,number2…)	=SUM(A1:A10)
Standard deviation, σ	STDEV(number1,number2…)	=STDEV(A1:A10)

Graph

To create a graph, select your numerical data to be graphed and choose the **Insert** tab, and pick **Scatter** from the **Charts** section. You will then need to select your graph and select the **Chart tools, Layout** tab. In this section you will see how to change your axis titles, graph labels, axis scales, include error bars and trendlines.



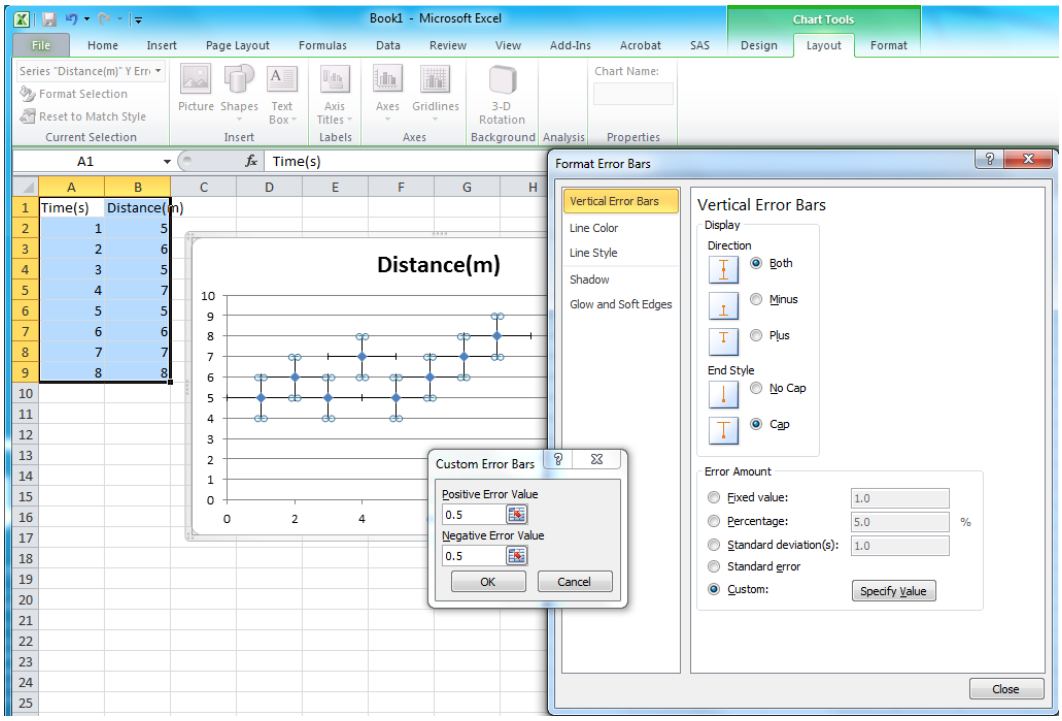
- **Scales -** Choose scales that make your graph simple and clear. Choose axis ranges that spread points evenly across the graph, rather than cramping them all into a corner. (This may mean that you need to exclude the zero point of an axis.)

- **Help -** If you are unsure of how to do something with your graph, explore the tab sections thoroughly and use Excel's **Help** function. Excel **can** do everything that you need – you only need to learn how (and where). If in doubt, ask a fellow student or your demonstrator.

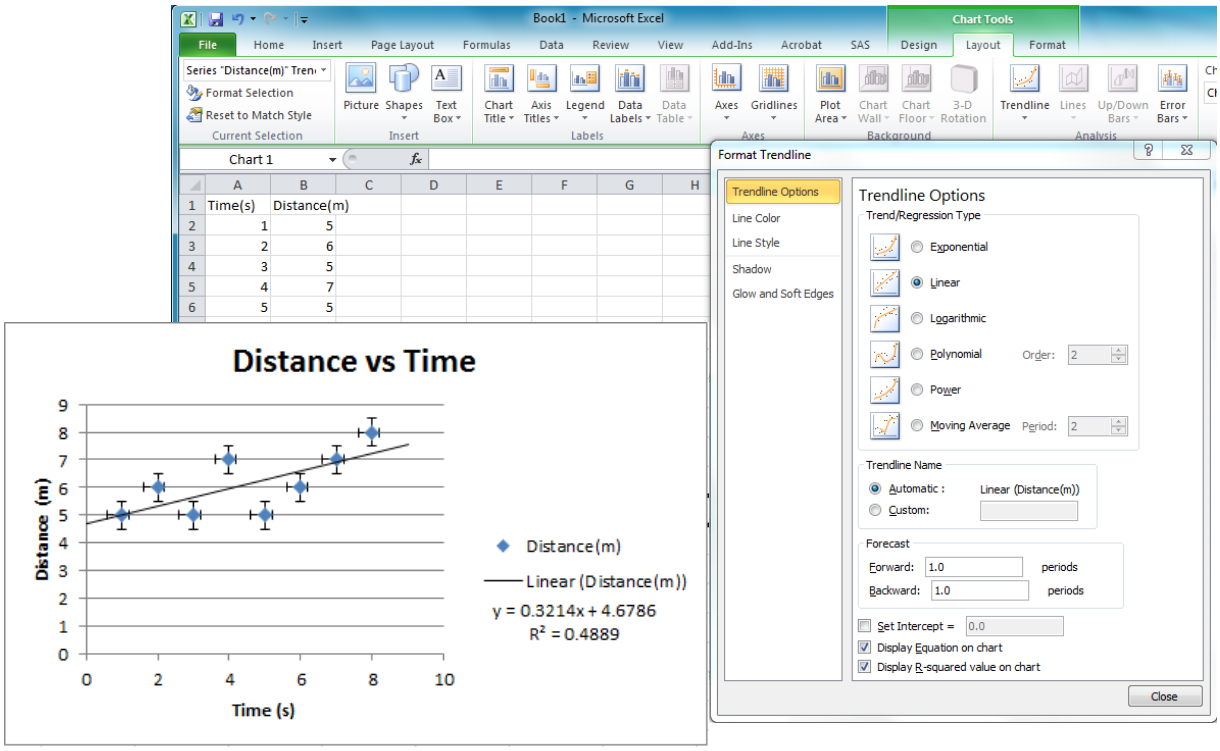
- **Changes -** When your graph is complete, remember that you can still alter it. You should right-click on elements of the graph to explore the options available.

Error bars in Excel -

- double-click the graph's data points,
- select the **Chart Tools, Layout** tab
- choose **Error bars**.
- select **more options,**
- use the fixed value to place your y error value or you can use **Custom Error bars**.
    - At this point Excel will also insert some default x error bars, if you have an error for x as well, select the horizontal error bar and insert correct value, or delete if you do not want them (sometimes they are so small you cannot see them to select – in which case do not worry about them).



Excel Trendlines



Excel can be made to construct lines of best fit for its graphs: select the data points on the graph, and right-click to choose **Add Trendline** from the pop-up menu. However, be careful: Excel's trendline methods won't work for all sets of data. Be sure to double-check that its lines look reasonable and sensible. Excel creates the Trendline via the Least Squares Fit method. We won't discuss this here, but you may like to investigate this method yourself.

You can make changes to the chart title, axes, legend, etc. by right clicking on the area or using the **Chart Tools, Layout** menu options. (see above figure)

Appendix C
SI Units

SI stands for *Le **S**ystème **I**nternational d'Unités*, which is the French title for the **International System of Units**. SI is the modern version of the metric system, and it is the system of measurements most widely used in science and commerce around the world today.

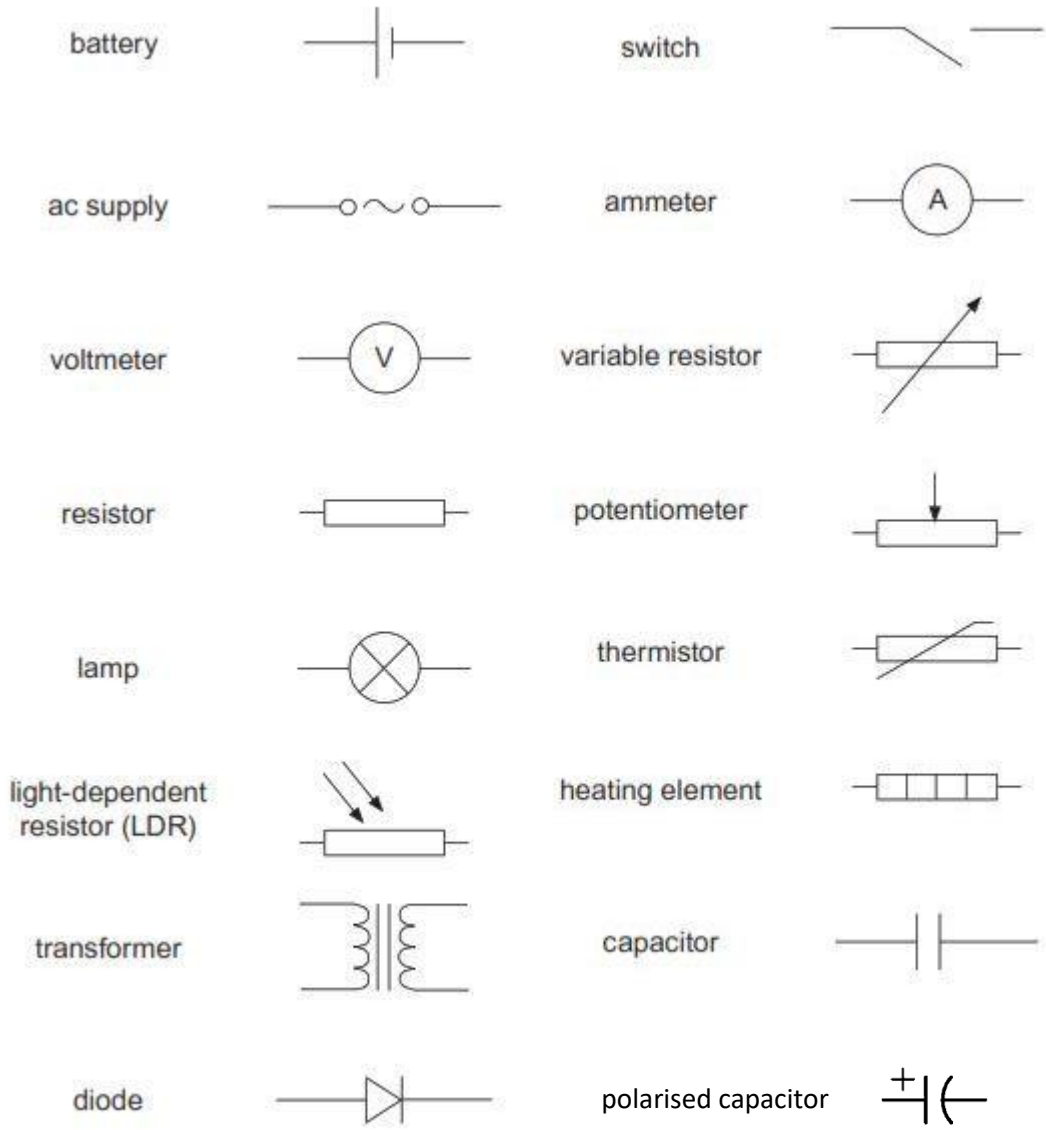
SI is based around seven **base units**: the metre **m** (length), the kilogram **kg** (mass), the second **s** (time), the ampere **A** (electric current), the kelvin **K** (temperature), the candela **cd** (luminous intensity), and the mole **mol** (amount of substance). These are the 'default' units for all measurements and physics equations. All other SI units of measurement (e.g., see table below) can be reduced to some combination of these seven base units.

Except in very special circumstances, all measurements and results should be made and quoted using SI units.

Quantity		**SI Unit**		
Name	**Symbol**	**Name**	**Symbol**	**SI Base Units**
Frequency	f, ν	Hertz	Hz	1 Hz = 1 / s
force	F	Newton	N	1 N = 1 kg m / s^2
energy	E	Joule	J	1 J = 1 N m
power	P	Watt	W	1 W = 1 J / s
electric charge	Q	Coulomb	C	1 C = 1 A s
electrical potential / 'potential difference' / 'electromotive force'	V, $\mathscr{E}$	Volt	V	1 V = 1 W / A = 1 kg m^2 / A s^3
electric capacitance	C	Farad	F	1 F = 1 A s / V = 1 A^2 s^4 / kg m^2
electric resistance	R	Ohm	Ω	1 Ω = 1 V / A = 1 kg m^2 / A^2 s^3
inductance	L	Henry	H	1 H = 1 V s / A = 1 kg m^2 / A^2 s^2

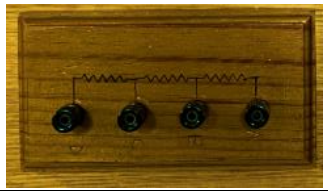
Appendix D
Electrical Circuit Components

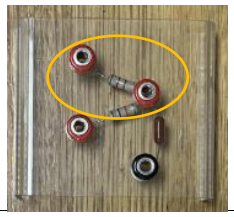
Symbols are used to represent electrical components in a circuit diagram to show how they are connected and to simplify complex circuits. Some common circuit components are shown below.



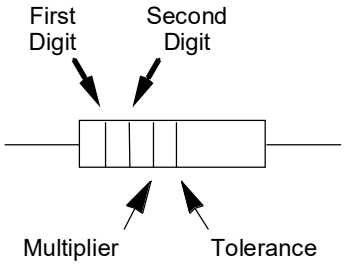
Resistors

Resistors are conductors, like wires, but they introduce a larger resistance to a circuit. Images of the type of resistors you use in the first year labs are:





Resistors are labelled in code according to their resistance. This code is indicated on the resistor by a set of coloured lines, as shown:



The first two colours represent the first two digits of the resistance value. The third colour represents the multiplication factor – the power of ten to multiply the digits by. The final colour represents the manufacturer's **tolerance** value (the level of uncertainty in the actual resistance).

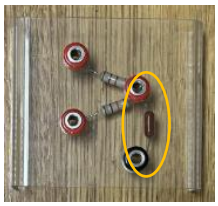
Colour	Number	Multiplier	Tolerance
Black	0	1	
Brown	1	10^1	1 %
Red	2	10^2	2 %
Orange	3	10^3	
Yellow	4	10^4	0.50 %
Green	5	10^5	0.25 %
Blue	6	10^6	0.10 %
Violet	7	10^7	0.05 %
Grey	8	10^8	
White	9	10^9	
Gold		10^{-1}	5 %
Silver		10^{-2}	10 %
No colour			20 %

For example: a resistor with the colour sequence Red-Green-Orange-Silver would have a resistance of 25,000 Ω ± 10%.

Capacitors

There are two types of capacitors; polarised and unpolarised. Polarized capacitors are also called electrolytic capacitors, they have a distinction between positive and negative terminals, unlike unpolarized capacitors. Safe operation of electrolytic capacitors requires the positive terminal to be at a higher voltage than the negative terminal. The polarities of a polarized capacitor are marked on the casing.

Images of the type of capacitors you use in the first year labs are:



unpolarised capacitors

polarised capacitors

Diodes

Diodes are made from semi-conductor materials, such as Silicon and Germanium. Unlike wires and resistors, they restrict the flow of current to one direction. The direction of current flow is shown by an arrow, or labelled as p-type and n-type, with current flowing from p to n end. An image of a type of diode you use in the first year labs is:

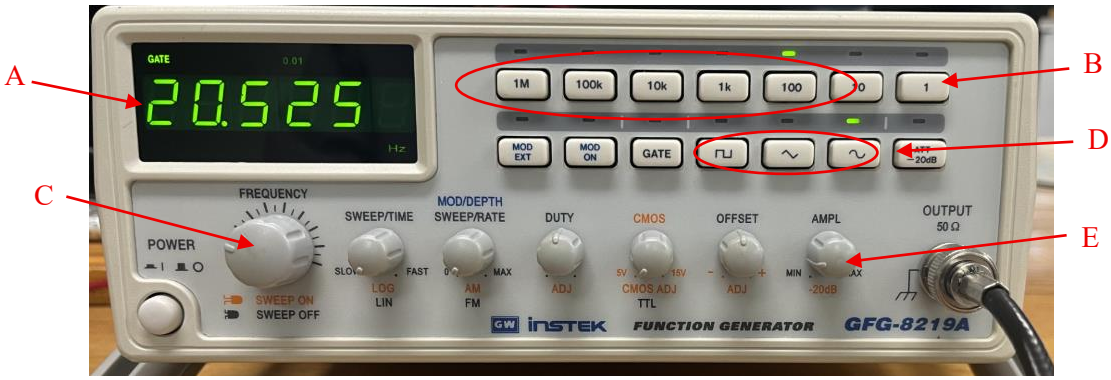
If you are interested, for more information on diodes refer to Appendix G.

Appendix E
Signal Generator and Oscilloscope

Signal Generator

Signal Generator outputs an AC voltage waveform that can be used as inputs to your circuits. In the labs there are two types of Signal Generators used. A picture of the front panels are shown below with the relevant controls highlighted and explained.

GFG-8219A Signal Generator



SFG-1013 Signal Generator

A – Displays the current output frequency, with the units shown in the bottom row.

B – Frequency range selection, used in conjunction with C to select a range of output frequencies.*

C – Frequency tuning knob. Turn this to vary the frequency by precise amounts.

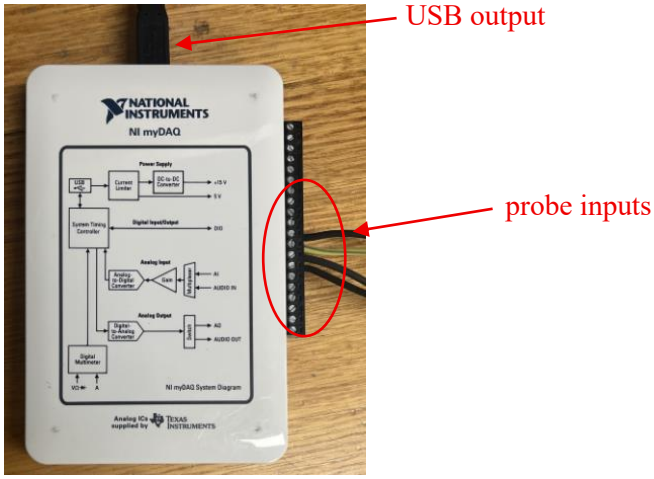
D – Wave shape selection. Choose between a square wave, triangle wave, and sine wave.

E – Amplitude selection. Changes the peak-to-peak voltage of the output waveform.

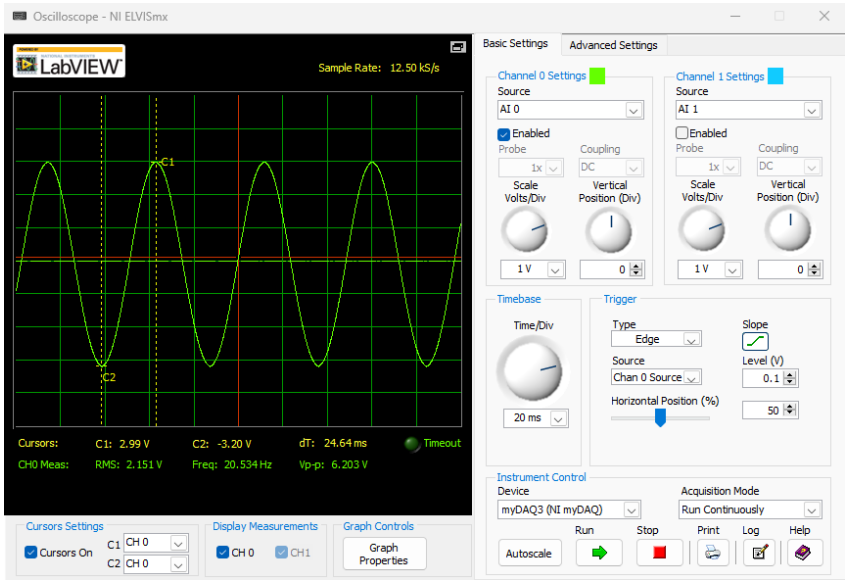
*For the SFG-1013, you will also need to use the shift button to choose the desired units then use output/ON to output the signal.

Digital Oscilloscope

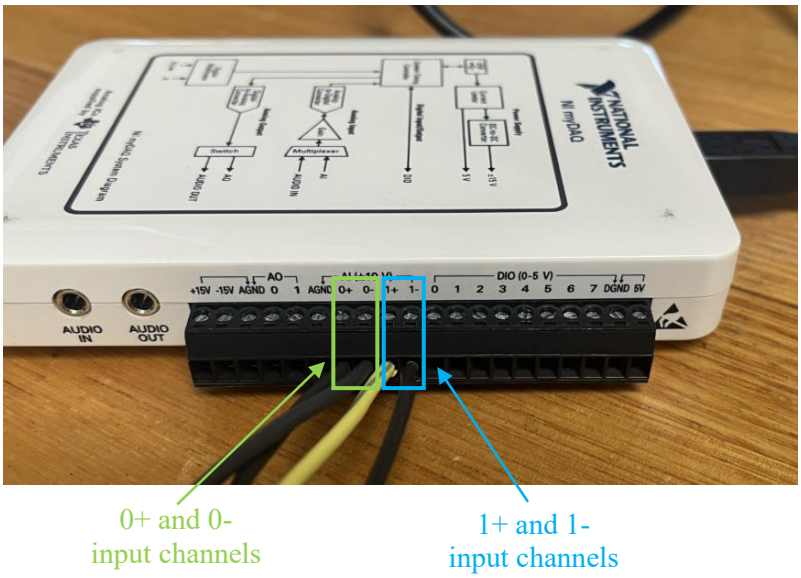
The digital Oscilloscope is used to visualize the voltage across components in a circuit over time. It is connected in a circuit in the same way as a voltmeter, i.e. parallel. The Oscilloscopes used in the labs are the NI ELVIS Oscilloscopes, which interface with the NI ELVISmx Oscilloscope software.



To use the Oscilloscope, connect the USB output to your computer. Start the NI ELVISmx oscilloscope software. You should see an interface like this:



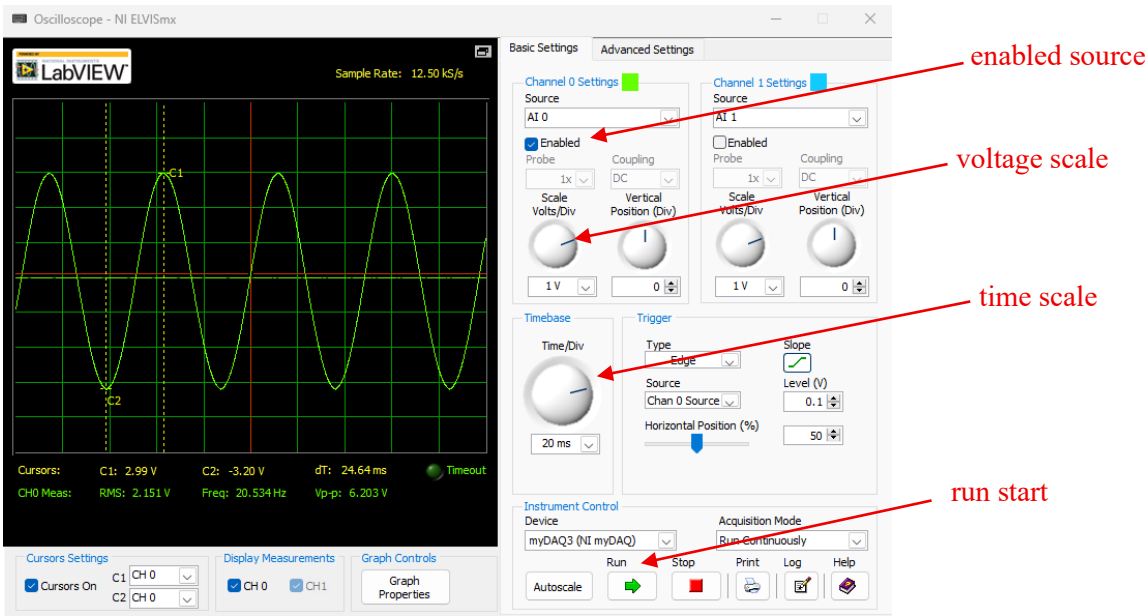
The Oscilloscope has two analog inputs in channels, meaning that two waveforms can be displayed simultaneously in the output graph. In the above image, two analog channels can be seen (AI 0 and AI 1), with only one channel enabled (AI 0) in this case. The two analog channels, AI 0 and AI 1 (in green and blue boxes respectively) correspond to the two probe pairs pictured below:



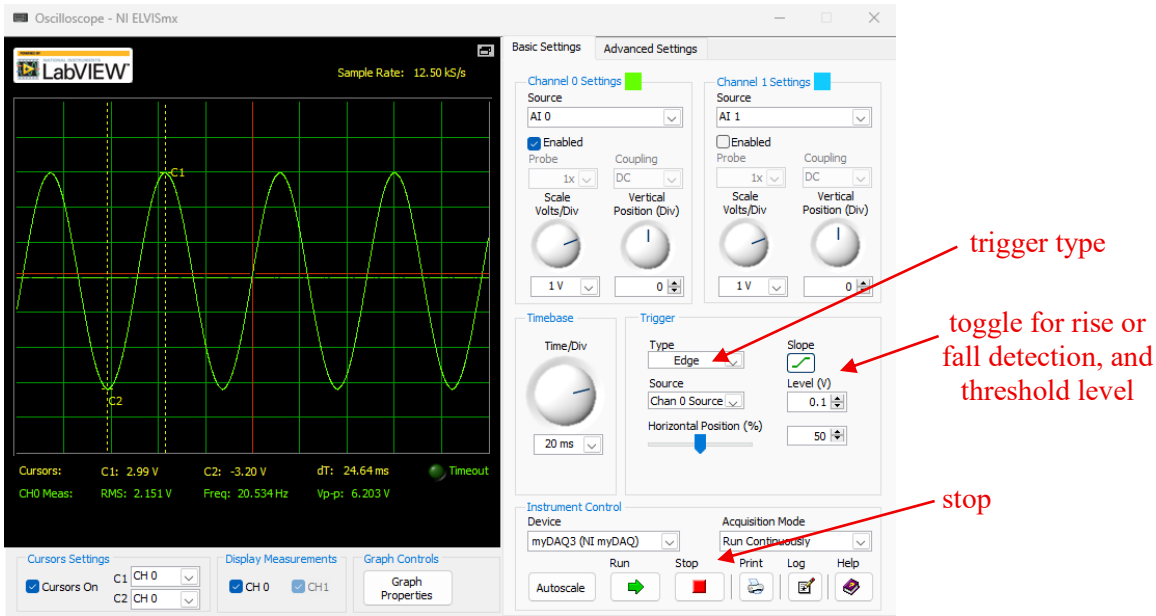
By connecting the 0+ and 0- leads (green box in image above) across your Signal Generator, for example, the input waveform can be seen in channel 0. The 1+ and 1- leads (blue box in the image above) can be connected across other component/s in your circuit, for example, the waveform across a capacitor.

Oscilloscope Software Settings

Enable the channel (source) you would to view on the graph and start the run. In the image below, source AI 0 is enabled, while source AI 1 is not enabled. To rescale the x axis of the graph, adjust the time division knob or manually enter a division scale value, as shown in the image below. To rescale the y axis of the graph, adjust the voltage division similarly.



The trigger will determine when the Oscilloscope starts the acquisition. Set the trigger type to Edge, this will start the acquisition when it has detected a rise (or fall) in the voltage above (or below) a set level. This ensures that the signal looks the same for different acquisition runs.



Once you obtain a suitable waveform, stop the acquisition to analyse the signal. On the bottom left (under graph), turn cursors on and drag the yellow dotted lines horizontally to make measurements at different times. “dT” shows the time interval between the two cursors. In the picture above, it is measuring the time between the valley of the wave and the peak of the wave. Make sure the two cursors are set on the correct channel.

Appendix F
Using the Falstad circuit simulator

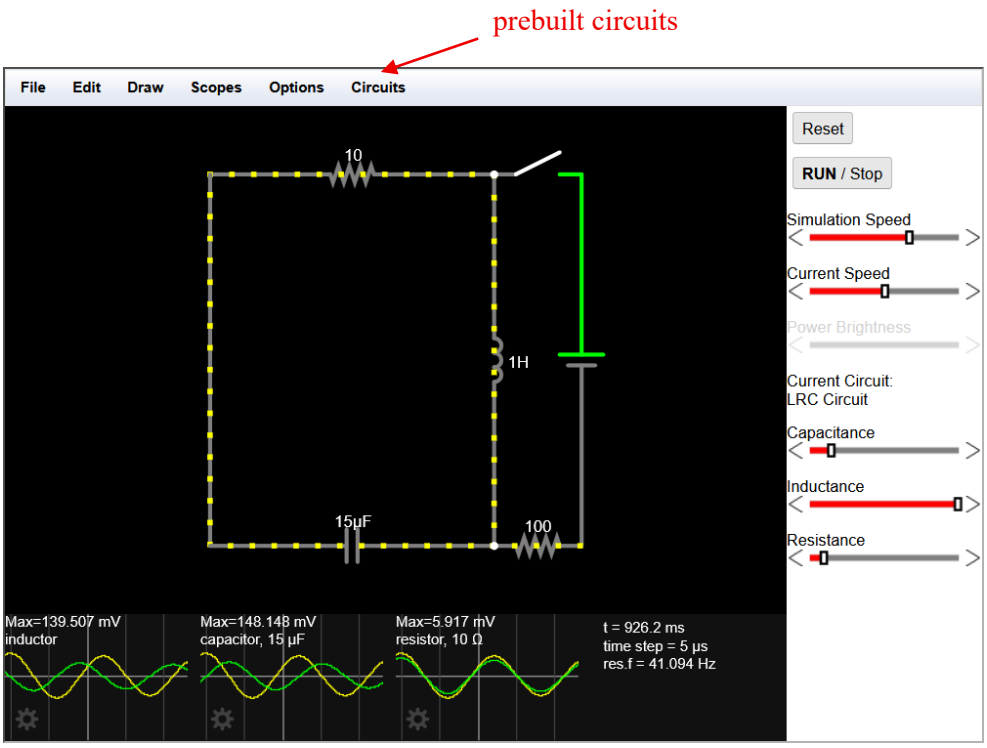
The Falstad circuit simulator is an interactive circuit sandbox applet that can be run within an internet browser. It can be used to build and simulate the flow of electricity through any circuit imaginable, including those seen in your labs.

You can access the browser applet here: https://www.falstad.com/circuit/

Detailed documentation of the features: https://www.falstad.com/circuit/doc/

Quick Start Guide

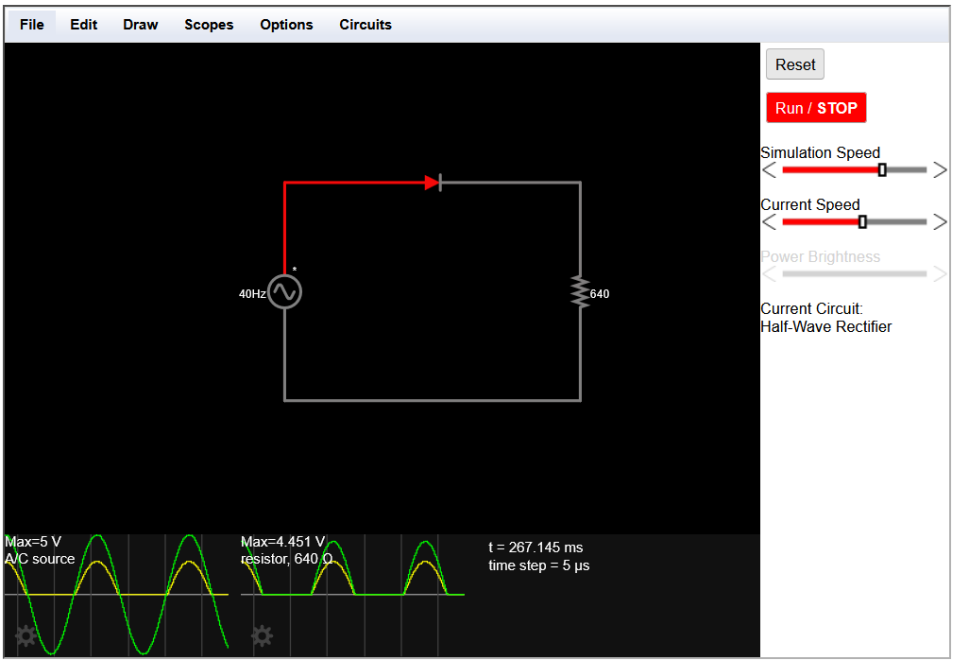
When you open the browser applet, a default circuit* will be running. Click on "Circuits" to choose a prebuilt circuit.



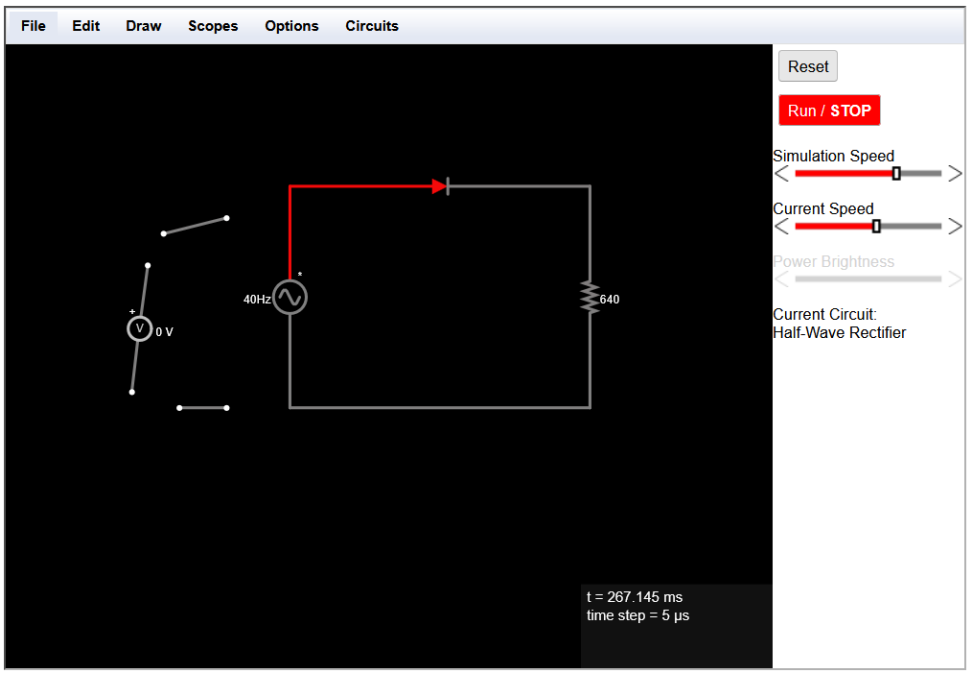
A pop-up menu will appear with categories of prebuilt circuits. As an example, a half-wave rectifier can be found under Diodes > Half-wave rectifier.

*Aside: the default circuit is a RLC circuit; a resistor (R), inductor (L) and capacitor (C) connected to a battery. This circuit is known as a harmonic oscillator and is used in many applications where tuning is required, for example traditional radio receivors and TV sets. It an also be used as low-pass or high-pass filter. The RLC circuit is described as a second-order circuit, and is mathematically descrived by a second-order differential equation in circuit analysis.

When you open the circuit, a schematic of the circuit will appear, and the circuit simulation will be running. In the image below, a half-wave rectifier is shown. The yellow dots move around the circuit, illustrating the flow of current, while the red and green colours represent positive and negative voltages, respectively. At the bottom of the window are traces that show the time-varying voltage and current that would be measured in different components. The image below shows a half-wave rectifier, with an AC source, resistor and a diode connected in series.

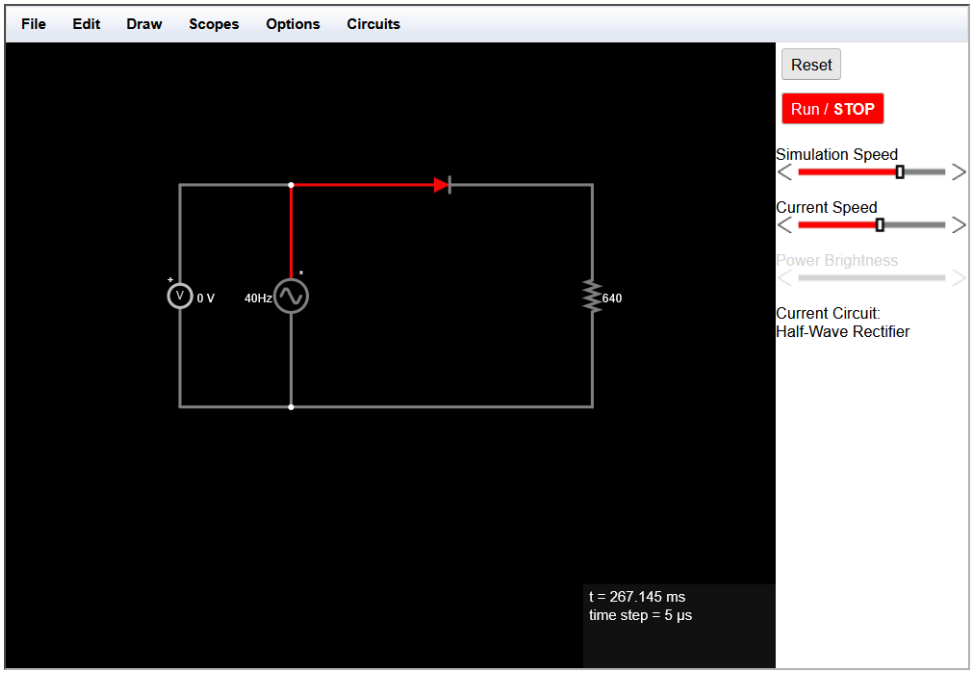


To set up your own probes of the voltage, mirroring what we do with Oscilloscopes in the DC Power lab, hover over the traces and hit “DELETE” to remove them. Use the “Draw” menu to select a voltmeter, found under Outputs and Labels > Add Voltmeter/Scope Probe. Place them in the schematic by clicking and dragging, as shown below. Press “w” to select a wire and draw two wires. Press “ESC” to go back to selection mode.

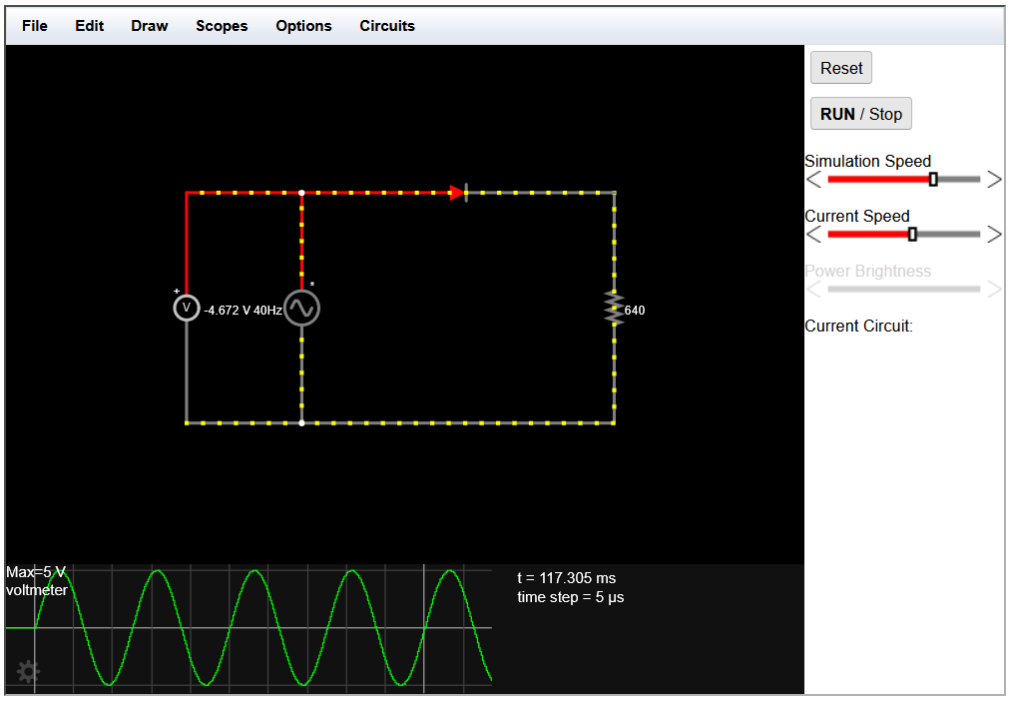


s

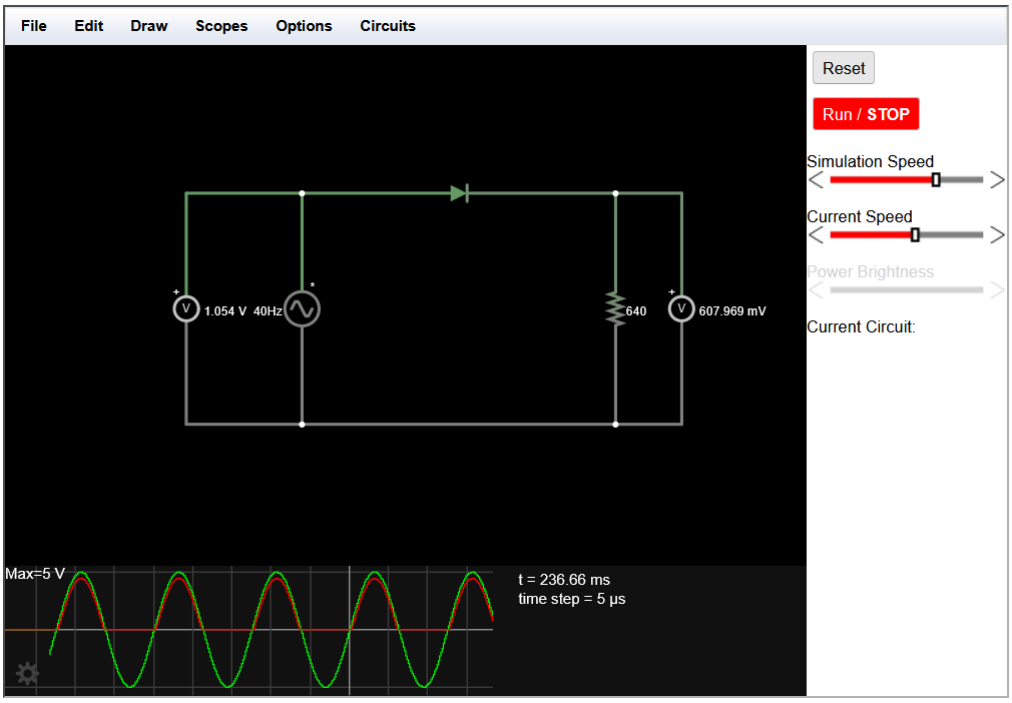
In select mode, connect the wires and scope probe by connecting the endpoints of the components.



After it is connected, hover over the voltmeter and right click, select "View in New Scope". When you run the simulation, you should see a sine voltage wave. This is what you would see in the experiment in your oscilloscope output.



Add another scope probe to the resistor. You can overlay the traces from the two probes by selecting "Add to Existing Scope". You can now compare the rectified output to the input, similar to what you do in the DC Power Supply lab with two Oscilloscope channels.



This circuit simulator is extremely powerful and flexible; this appendix covers only the bare minimum features pertaining to the labs. As an example of what it can do, here is a simulation of the game "Pong" implemented using this software https://www.falstad.com/pong/vonly.html.

Appendix G
More about Diodes

Silicon and Germanium are semiconductor materials commonly used to construct diodes and transistors. Like Carbon, these elements have a chemical valence number of **four** – this means that they form chemical bonds by sharing their four valence electrons with neighbouring atoms. At absolute zero all these covalent chemical bonds are intact, but at room temperature some bonds will fracture due to thermal vibrations. As a result, some electrons will be able to move freely through the molecular crystal lattice.

In addition to the freely moving electrons, there will also be **holes** in the lattice where the electrons have escaped. When a neighbouring electron leaves its own position to fill one of these holes, a new hole will be created in the neighbouring position. In this way the holes will appear to 'move' through the lattice, just as the extra free electrons may move around. Equal numbers of freely-moving electrons and holes therefore act as effective carriers of negative and positive charge through the semiconductor. (The extra energy needed to create an electron-hole pair is 1.1 eV for Silicon and only 0.7 eV for Germanium.)

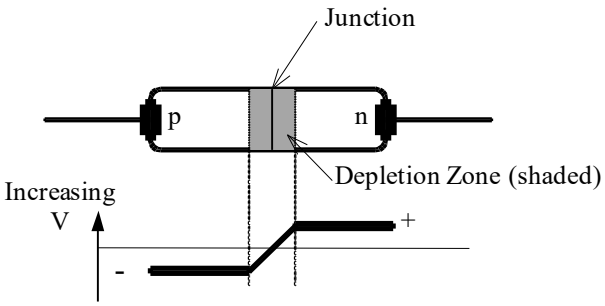
If the semiconductor is **doped** – by adding a controlled amount of certain impurities to its chemical structure – then the balance between positive and negative charge carriers (the electrons and holes) can be upset, and the conductivity of the semiconductor can increase dramatically. For example: if an element of valence **five** (e.g. Phosphorus, Arsenic) is added, this impurity will try to fit into the semiconductor lattice. But at the impurity site there will now be an extra valence electron, free to move through the lattice. Even if the amount of impurity added is only 1 part in 10^8, the conductivity may increase by a factor of more than 10.

The impurity atom will be positively charged, but it cannot accommodate any neighbouring free electrons (four bonds are already filled), so it will not behave like a usual semiconductor 'hole'. This type of doped semiconductor is called **n-type** (for 'negative'), because its electrical current will be mostly carried by electrons (they will be the 'majority carriers'). The impurity atoms of valence five are called **donor impurities**, because they 'donate' electrons to the conduction process.

Similarly, one can instead add **acceptor impurities** of valence **three** (e.g. Aluminium, Gallium, Indium), which will accept electrons at the impurity sites and thereby create extra 'holes' in the molecular lattice. These are called **p-type** semiconductors, because their majority carriers are the positively charged holes.

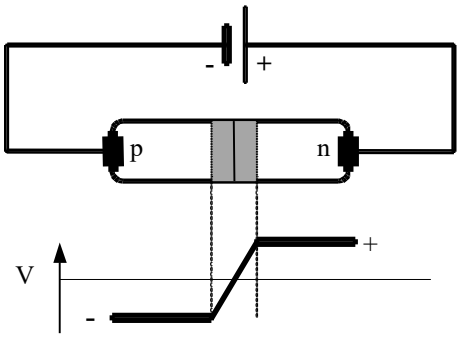
The Junction Diode

When p-type and n-type regions are **both** present within the same semiconductor crystal, the place where they meet forms a semiconductor **junction diode**. Mobile electrons from the n-type region and holes from the p-type region will move towards each other, combining to create a **depletion zone** where the charges have cancelled each other out. But this flow of current will not continue forever – eventually, holes moving towards the n-type region will be repelled by the electric field created by the holes that have already moved there, and electrons moving towards the p-type region will be repelled by the electric field of electrons that have already moved there.



The p-type region is left with an overall negative charge, and the n-type region is left with an overall positive charge. The net result is a potential difference (voltage) across the depletion zone (the region with no mobile carriers).

Now consider what happens when an external source of voltage (e.g. a battery) is connected across the diode.



If the voltage increases across the diode, then no net number of majority carriers will be able to cross the junction (the depletion zone) and almost no current will flow. Only the small number of thermally released 'minority carriers' (holes in the n-type region and electrons in the p-type region) will be able to cross the junction and create a current. (The current formed by minority carriers in this way is typically very small – microamps at most.)

However: if the voltage is gradually applied with opposite polarity to that shown above, then the internal potential difference (voltage) will gradually decrease and then reverse. Majority carriers on either side would then be able to cross the junction – a relatively large current would be established. As the voltage increases, the resistance of the diode drops to near zero and the current increases rapidly.

In this way, the semiconductor chemistry of the diode effectively allows current to flow in one direction but not the other – an extremely useful electronic property.